

Technology #173

Integrate a pre-engineered microfracture lattice into the drainage floor surface, creating a passive network of channels that open under hydraulic pressure to compensate for viscosity-induced resistance.

Engineering Technology-Transfer Dossier

CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION

Generated: 2026-09-02

Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

Integrate a pre-engineered microfracture lattice into the drainage floor surface, creating a passive network of channels that open under hydraulic pressure to compensate for viscosity-induced resistance.

WHY DOES IT MATTER?

Documented problem 'hydraulic performance deviation' concerning drainage floor. User-submitted engineering problem (session input): "Why does P-07's drainage floor deviate from the measured water viscosity basis? (reality discrepancy semantics)". [fra_rail_accidents] retrieval status GRAMMAR_MISMATCH (query does not match source grammar: YYYY-MM-DD date floor (\$where date >= X) — engine routing error, not a provider failure (not absence, Art. XXV)) — provider outcome recorded, not treated as absence (Art. XXI.3). [doe_osti] 10 records retrieved live (2026-2026). Illustrative records: 'Evaluating the Analytical Capabilities of RAM In-situ Diagnostics'; 'Resonant Acoustic Mixing of Dense Pastes: Role of Adhesion and Viscosity'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 3 governing equations, 13 design inputs, 6 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in fluidics_hydraulic. 10-step build plan with specific test articles and acceptance criteria. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHAT DOES THE BUYER GET?

Buyer receives: canonical invention specification (with epistemic classes).; Buyer receives: engineering specification (this document, all blocks classed).; Buyer receives: design input/output graph with explicit IDs.; Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).; Buyer receives: killer-experiment definition and its EIG basis.; Buyer receives: evidence custody references and hashes.; Buyer receives: unresolved-uncertainty register.; Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

WHAT DOES THE BUYER HAVE TO BUILD?

Buyer must create: actual geometry and detailed design.; Buyer must create: sourced engineering parameters and acceptance thresholds.; Buyer must create: physical prototypes.; Buyer must create: all verification and validation results.; Buyer must create: manufacturing process qualification.; Buyer must create: regulatory pathway determination and submissions.

WHAT IS THE NEXT DECISIVE EXPERIMENT?

first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

WHAT WOULD MAKE US KILL IT?

killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.4557178723356231, EIG-per-cost 0.4557178723356231; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 3 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 10-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed) - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: physical bench/clinical experiment — NOT simulation. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - article for VF-001 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: Perform flow■rate measurements on a prototype floor with the micro■fracture lattice while varying water viscosity (e.g., by adding glycer.... Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - article for VF-003 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: bench flow-loop characterization across operating range. Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 3 equations. Key assumptions: CSF approximated as Newtonian (valid at shunt flow rates); no wall compliance effects; transition criterion $Re \sim 2300$ is a MODEL_DERIVED engineering rule of thumb; $C_d \sim 0.6-0.65$ is an engineering prior, not a measurement; FLUID-001 (Hagen-Poiseuille (laminar pipe flow)) is applicable only under: Newtonian fluid, laminar regime, rigid circular lumen, fully developed flow.; FLUID-002 (Reynolds number) is applicable only under: circular section; single-phase flow.; FLUID-003 (Orifice flow (sharp-edged restriction)) is applicable only under: turbulent orifice flow; C_d treated as MODEL_DERIVED until measured.; Model input variables requiring sourced values before any numeric result: r (m), dP (Pa), μ (Pa.s), L (m), ρ (kg/m³), v (m/s), D (m), μ (Pa.s), C_d (-), A (m²), dP (Pa), ρ (kg/m³) — all UNKNOWN (no sourced value exists; Art. VIII).; Model output variables produced by the governing model: Q (m³/s), Re (-), Q (m³/s) — UNKNOWN until the inputs are sourced and the model is evaluated.; Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): lumen inner diameter [r , m] value_status=UNKNOWN; lumen length [r , m] value_status=UNKNOWN; surface roughness [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; operating pressure head [dP , Pa] value_status=UNKNOWN; flow rate range [Q , m³/s] value_status=UNKNOWN; viscosity of flowed medium [μ , Pa.s] value_status=UNKNOWN; occlusion growth tolerance [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

3 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: Pore■fracture structure provides passive hydraulic pathways that modulate flow resistance.

POTENTIAL VALUE DRIVER: Addresses: Documented problem 'hydraulic performance deviation' concerning drainage floor. User-submitted engineering problem (session input): "Why does P-07's drainage floor deviate from the measured water viscosity basis? (reality discrepancy semantics)". [fra_rail_accidents] retrieval status GRAMMAR_MISMATCH (query does not match source grammar: YYYY-MM-DD date floor (\$where date >= X) — engine routing error, not a provider failure (not absence, Art. XXV)) — provider outcome recorded, not treated as absence (Art. XXI.3). [doe_osti] 10 records retrieved live (2026-2026). Illustrative records: 'Evaluating the Analytical Capabilities of RAM In-situ Diagnostics'; 'Resonant Acoustic Mixing of Dense Pastes: Role of Adhesion and Viscosity'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: domain-appropriate technology buyer (to be identified by portfolio owner)

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: fluidics_hydraulic

Disciplines: fluid mechanics, mechanical engineering, biomedical engineering, materials science

Architecture: PROPOSED architecture (no physical system exists): flow path / lumen architecture; pressure regulation element; sensing / feedback element (if closed-loop); termination interfaces (patient / reservoir) — realizing: Integrate a pre-engineered microfracture lattice into the drainage floor surface, creating a passive network of channels that open under hydraulic pressure to compensate for viscosity-induced resistance.

Subsystems:

?: flow path / lumen architecture - ?

?: pressure regulation element - ?

?: sensing / feedback element (if closed-loop) - ?

?: termination interfaces (patient / reservoir) - ?

2. Mechanism Architecture

Physical Changes: PROPOSED: Integrate a pre-engineered microfracture lattice into the drainage floor surface, creating a passive network of channels that open under hydraulic pressure to compensate for viscosity-induced resistance.

Key Physics: Fluid mechanics / hydraulics governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

3. Governing Engineering Model

Summary: Fluid mechanics / hydraulics governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Equations:

FLUID-001: Hagen-Poiseuille (laminar pipe flow). Expression: $Q = (\pi * r^4 * dP) / (8 * \mu * L)$.

Applicability to this invention: APPLICABLE; holds under: Newtonian fluid, laminar regime, rigid circular lumen, fully developed flow. Assumptions: CSF approximated as Newtonian (valid at shunt flow rates); no wall compliance effects. Engaged variables: dP , μ .

FLUID-002: Reynolds number. Expression: $Re = (\rho * v * D) / \mu$. Applicability to this invention: APPLICABLE; holds under: circular section; single-phase flow. Assumptions: transition criterion $Re \sim 2300$ is a MODEL_DERIVED engineering rule of thumb. Engaged variables: D , μ .

FLUID-003: Orifice flow (sharp-edged restriction). Expression: $Q = C_d * A * \sqrt{2 * dP / \rho}$. Applicability to this invention: APPLICABLE; holds under: turbulent orifice flow; C_d treated as MODEL_DERIVED until measured. Assumptions: $C_d \sim 0.6-0.65$ is an engineering prior, not a measurement. Engaged variables: dP .

Assumptions:

- CSF approximated as Newtonian (valid at shunt flow rates)
- no wall compliance effects
- transition criterion $Re \sim 2300$ is a MODEL_DERIVED engineering rule of thumb
- $C_d \sim 0.6-0.65$ is an engineering prior, not a measurement
- FLUID-001 (Hagen-Poiseuille (laminar pipe flow)) is applicable only under: Newtonian fluid, laminar regime, rigid circular lumen, fully developed flow.
- FLUID-002 (Reynolds number) is applicable only under: circular section; single-phase flow.
- FLUID-003 (Orifice flow (sharp-edged restriction)) is applicable only under: turbulent orifice flow; C_d treated as MODEL_DERIVED until measured.

- Model input variables requiring sourced values before any numeric result: r (m), dP (Pa), μ (Pa.s), L (m), ρ (kg/m³), v (m/s), D (m), μ (Pa.s), C_d (-), A (m²), dP (Pa), ρ (kg/m³) — all UNKNOWN (no sourced value exists; Art. VIII).
- Model output variables produced by the governing model: Q (m³/s), Re (-), Q (m³/s) — UNKNOWN until the inputs are sourced and the model is evaluated.
- Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): lumen inner diameter [r , m] value_status=UNKNOWN; lumen length [r , m] value_status=UNKNOWN; surface roughness [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; operating pressure head [dP , Pa] value_status=UNKNOWN; flow rate range [Q , m³/s] value_status=UNKNOWN; viscosity of flowed medium [μ , Pa.s] value_status=UNKNOWN; occlusion growth tolerance [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.

Boundary Conditions:

- to be established by the first bench characterization (not yet performed)

Failure Regimes:

- {'regime': 'hydraulic performance deviation', 'basis': 'SOURCE_FACT (problem.json)'}
- {'regime': 'proximal occlusion by tissue ingrowth', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'lumen collapse under external pressure (kink)', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'encrustation / mineral deposition', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'regime transition to turbulence altering calibration', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'check-valve seat wear opening reverse leakage', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}

4. Design Inputs

ID	Input	Value	Origin
DI-001	Stated constraint	maintain design flow rate across viscosity range	SOURCE_FACT
DI-002	Mechanism driver (proposed)	Pore ■ fracture structure provides passive hydraulic pathways that modulate MODELS force.	MODELS
DI-003	Expected effect target	The passive micro ■ fracture network maintains the design flow rate across the MODELS viscosity range, reducing the hydraulic	MODELS
DI-004	Domain design input pattern: required flow applicability: variable pressure [pressure]	required flow applicability: variable pressure [pressure]	ENGINEERING_PROPOSED
DI-005	Domain design input pattern: medium viscosity applicability: variable viscosity [viscosity]	medium viscosity applicability: variable viscosity [viscosity]	ENGINEERING_PROPOSED
DI-006	Domain design input pattern: anticipated applicability: NOT_ESTABLISHED (candidate pattern; confirm against ENGINEERING_PROPOSED)	anticipated applicability: NOT_ESTABLISHED (candidate pattern; confirm against ENGINEERING_PROPOSED)	ENGINEERING_PROPOSED
DI-007	Domain design input pattern: device lifetime applicability: NOT_ESTABLISHED (candidate pattern; confirm against ENGINEERING_PROPOSED)	device lifetime applicability: NOT_ESTABLISHED (candidate pattern; confirm against ENGINEERING_PROPOSED)	ENGINEERING_PROPOSED
DI-008	Regulatory design input: compliance candidate standards (e.g. ISO 13485 (QMS) and further, pathway to EN ISO 13485 (QMS) for CE marking)	compliance candidate standards (e.g. ISO 13485 (QMS) and further, pathway to EN ISO 13485 (QMS) for CE marking)	ENGINEERING_PROPOSED
DI-009	Custodied observation: Design, constraints: https://hydroplation.org/articles/9234344 , metadata: https://hydroplation.org/articles/9234344 , metadata: https://hydroplation.org/articles/9234344	constraints: https://hydroplation.org/articles/9234344 , metadata: https://hydroplation.org/articles/9234344 , metadata: https://hydroplation.org/articles/9234344	ENGINEERING_PROPOSED
DI-010	Custodied observation: Ensemble https://ukrpeis.org/stability/41518662 , and optimization of https://ukrpeis.org/stability/41518662	Ensemble https://ukrpeis.org/stability/41518662 , and optimization of https://ukrpeis.org/stability/41518662	ENGINEERING_PROPOSED
DI-011	Custodied observation: Study on arxiv.org/abs/1608.02118 , posttest_bashall.html, https://arxiv.org/abs/1608.02118	Study on arxiv.org/abs/1608.02118 , posttest_bashall.html, https://arxiv.org/abs/1608.02118	ENGINEERING_PROPOSED
DI-012	Custodied observation: Review on arxiv.org/abs/1608.02118 , Technical https://arxiv.org/abs/1608.02118	Review on arxiv.org/abs/1608.02118 , Technical https://arxiv.org/abs/1608.02118	ENGINEERING_PROPOSED
DI-013	Custodied observation: Mapping https://arxiv.org/abs/1608.02118 , https://arxiv.org/abs/1608.02118	Mapping https://arxiv.org/abs/1608.02118 , https://arxiv.org/abs/1608.02118	ENGINEERING_PROPOSED

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	flow path / lumen architecture realizing Integrate a pre ■ engineered micro ■ fracture lattice into the drainage floor surface, creating a passive network of channels	ENGINEERING_PROPOSED	details of the micro ■ fracture lattice structure
DO-002	pressure regulation element realizing Integrate a pre ■ engineered micro ■ fracture lattice into the drainage floor surface, creating a passive network of channels	ENGINEERING_PROPOSED	details of the micro ■ fracture lattice structure
DO-003	sensing / feedback element (if closed-loop) realizing Integrate a pre ■ engineered micro ■ fracture lattice into the drainage floor surface, creating a passive network of channels	ENGINEERING_PROPOSED	details of the micro ■ fracture lattice structure
DO-004	termination interfaces (patient / reservoir) realizing Integrate a pre ■ engineered micro ■ fracture lattice into the drainage floor surface, creating a passive network of channels	ENGINEERING_PROPOSED	details of the micro ■ fracture lattice structure
DO-005	flow path / lumen architecture interfaces with pressure regulation element: signal transfer material (thin film) (if dimensions to be specified) engineered micro ■ fracture lattice	ENGINEERING_PROPOSED	details of the micro ■ fracture lattice structure
DO-006	pressure regulation element interfaces with sensing / feedback element (if closed-loop): signal transfer material (thin film) (if dimensions to be specified) engineered micro ■ fracture lattice	ENGINEERING_PROPOSED	details of the micro ■ fracture lattice structure
DO-007	sensing / feedback element (if closed-loop) interfaces with termination interfaces (patient / reservoir): signal transfer material (thin film) (if dimensions to be specified) engineered micro ■ fracture lattice	ENGINEERING_PROPOSED	details of the micro ■ fracture lattice structure
DO-008	Device boundaries: patient/fluid inlet and outlet termination control/data processing required by telemedicine, seal ■ engineered micro ■ fracture lattice	ENGINEERING_PROPOSED	details of the micro ■ fracture lattice structure
DO-009	Flow/force/signal path geometry realizing the mechanism of a pre ■ engineered micro ■ fracture lattice, surface of the drainage floor surface, creating a passive network of channels	ENGINEERING_PROPOSED	details of the micro ■ fracture lattice structure
DO-010	Operating range for lumen inner diameter within which the mechanism holds its stated behavior or computed bounds for lumen inner diameter; verification of range endpoints	ENGINEERING_PROPOSED	details of the micro ■ fracture lattice structure
DO-011	Operating range for lumen length within which the mechanism holds its stated behavior or computed bounds for lumen length; verification of range endpoints	ENGINEERING_PROPOSED	details of the micro ■ fracture lattice structure
DO-012	Operating range for surface roughness within which the mechanism holds its stated behavior or computed bounds for surface roughness; verification of range endpoints	ENGINEERING_PROPOSED	details of the micro ■ fracture lattice structure
DO-013	Operating range for operating pressure head within which the mechanism holds its stated behavior or computed bounds for operating pressure head; verification of range endpoints	ENGINEERING_PROPOSED	details of the micro ■ fracture lattice structure
DO-014	Operating range for flow rate range within which the mechanism holds its stated behavior or computed bounds for flow rate range; verification of range endpoints	ENGINEERING_PROPOSED	details of the micro ■ fracture lattice structure
DO-015	Operating range for viscosity of flowed medium within which the mechanism holds its stated behavior or computed bounds for viscosity of flowed medium; verification of range endpoints	ENGINEERING_PROPOSED	details of the micro ■ fracture lattice structure
DO-016	No closed-loop control is proposed: the invention operates as an open-loop as specified; this record exists so the omission is explicit and auditable	ENGINEERING_PROPOSED	details of the micro ■ fracture lattice structure
DO-017	Information path: sensing element -> decision/monitor control/data processing required by telemedicine, seal ■ engineered micro ■ fracture lattice	ENGINEERING_PROPOSED	details of the micro ■ fracture lattice structure
DO-018	Fixture enabling: bench flow-loop characterization across the operating range — rig https://arxiv.org/abs/1608.02118 , build specification https://arxiv.org/abs/1608.02118	ENGINEERING_PROPOSED	details of the micro ■ fracture lattice structure
DO-019	Fixture enabling: pressure-drop mapping vs flow (identifying flow transitions) — rig https://arxiv.org/abs/1608.02118 , build specification https://arxiv.org/abs/1608.02118	ENGINEERING_PROPOSED	details of the micro ■ fracture lattice structure

ID	Description	Status	Missing Inputs
DO-020	Fixture enabling: accelerated occlusion / fouling challenge	UNKNOWN — rig concept finalized, build type, fabrication UNKNOWN	None; acceptance instrumentation
DO-021	Fixture enabling: pressure-flow mapping across full operating envelope	UNKNOWN — rig concept generated, build type, fabrication UNKNOWN	None; acceptance instrumentation

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
lumen inner diameter	UNKNOWN (no sourced value)	m	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
lumen length	UNKNOWN (no sourced value)	m	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
surface roughness	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
operating pressure head	UNKNOWN (no sourced value)	Pa	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
flow rate range	UNKNOWN (no sourced value)	m^3/s	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
viscosity of flowed medium	UNKNOWN (no sourced value)	Pa.s	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
occlusion growth tolerance	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
hydraulic performance deviation	Failure of hydraulic performance design	deviation control	the QLE-VIS7-BUS-HB2 (re)validates design	UNKNOWN (no sourced value) while the operating condition state (Attr.)
proximal occlusion by tissue ingrowth	Failure of proximal occlusion by tissue ingrowth	by tissue ingrowth self-polishing	the QLE-VIS7-BUS-HB2 (re)validates design	UNKNOWN (no sourced value) while the operating condition state (Attr.)
lumen collapse under external pressure	Failure of lumen collapse under external pressure	external pressure (kPa) external pressure bending	the QLE-VIS7-BUS-HB2 (re)validates design	UNKNOWN (no sourced value) while the operating condition state (Attr.)
encrustation / mineral deposition	Failure of encrustation / mineral deposition	encrustation / mineral deposition	the QLE-VIS7-BUS-HB2 (re)validates design	UNKNOWN (no sourced value) while the operating condition state (Attr.)
regime transition to turbulence	Failure of regime transition to turbulence	regime transition to turbulence	the QLE-VIS7-BUS-HB2 (re)validates design	UNKNOWN (no sourced value) while the operating condition state (Attr.)
check-valve seat wear	Failure of check-valve seat wear	check-valve seat wear	the QLE-VIS7-BUS-HB2 (re)validates design	UNKNOWN (no sourced value) while the operating condition state (Attr.)

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
hydraulic performance deviation	Failure by hydraulic performance deviation	design device of NOT ESTABLISHED	residual uncertainty condition (A and X)
proximal occlusion by tissue ingrowth	Failure by proximal occlusion by tissue ingrowth	integrated struts surface dissoluble	high level of drug growths that candidate fails the free-flow test (A and X)
lumen collapse under external pressure (kink)	Failure by pressure (collapse under external pressure)	wall thickness and external pressure monitoring	thin wall thickness is expected to inhibit lumen collapse (A and X)
encrustation / mineral deposition	Failure by encrustation / mineral deposition	surface electrolyte coating selection	the future study when zones of drug delivery are targeted (A and X)
regime transition to turbulence	Failure by regime transition to turbulence	altering geometry above the regime	high NOA fluid in the subsonic regime failure depends (A and X)
check-valve seat wear opening	Failure by check-valve seat wear opening	seal material and geometry; seal life	high NOA valve seat material is using failure depends (A and X)

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
VF-001	Perform flow-rate measurements on a prototype	flow rate measurement	indicated during the test	the flow rate is within the target range
VF-002	run killer experiment: falsification_test	kill rate test	kill rate test	kill rate test
VF-003	bench flow-loop characterization across operating range	Acceptance criteria	Acceptance criteria	Acceptance criteria
VF-004	pressure-drop mapping vs flow (identify pressure drops)	Acceptance criteria	Acceptance criteria	Acceptance criteria
VF-005	delivered-flow trend below baseline in adverse conditions	delivered-flow trend	delivered-flow trend	delivered-flow trend
VF-006	kink/collapse test across duty cycle; flow kink/collapse test	kink/collapse test	kink/collapse test	kink/collapse test
VF-007	pressure-flow mapping across the full operating range	pressure-flow mapping	pressure-flow mapping	pressure-flow mapping
VF-008	reverse-leakage rate measurement after accelerated cycling	reverse-leakage rate measurement	reverse-leakage rate measurement	reverse-leakage rate measurement

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VA-001	end-use validation of verified design	NOT ESTABLISHED (defined after verification)	acceptance criteria to be fixed	Validation is in progress

11. Materials

Component	Candidate Material	Source	Verification Required	Status
domain candidate material	medical-grade silicone elastomer	standard CSF shunt catheter material	True	ENGINEERING_PROPOSED
domain candidate material	polycarbonate / PSU rigid components	standard shunt valve housings	True	ENGINEERING_PROPOSED

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
1	medical-grade silicone elastomer	1	Material	medical-grade silicone elastomer	NOT ESTABLISHED	NOT ESTABLISHED	Applicability verification required
2	polycarbonate / PSU rigid components	1	Material	polycarbonate / PSU rigid components	NOT ESTABLISHED	NOT ESTABLISHED	Applicability verification required

13. Manufacturing

Status: NO MANUFACTURING ANALYSIS POSSIBLE — no design exists; processes are candidates only

Candidate Processes:

- medical extrusion with inner-diameter tolerance control: domain registry manufacturing route - candidate only; no process qualification exists
- injection molding of valve/housing features: domain registry manufacturing route - candidate only; no process qualification exists
- laser drilling of calibrated orifices: domain registry manufacturing route - candidate only; no process qualification exists

14. External Evidence

Source 1:

URL :

Excerpt:

Source 2:

URL :

Excerpt:

Source 3:

URL :

Excerpt:

15. Transfer Boundary

YOU RECEIVE:

- + Buyer receives: canonical invention specification (with epistemic classes).
- + Buyer receives: engineering specification (this document, all blocks classed).
- + Buyer receives: design input/output graph with explicit IDs.
- + Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).
- + Buyer receives: killer-experiment definition and its EIG basis.
- + Buyer receives: evidence custody references and hashes.
- + Buyer receives: unresolved-uncertainty register.
- + Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

YOU MUST DEVELOP:

- > Buyer must create: actual geometry and detailed design.
- > Buyer must create: sourced engineering parameters and acceptance thresholds.
- > Buyer must create: physical prototypes.
- > Buyer must create: all verification and validation results.
- > Buyer must create: manufacturing process qualification.
- > Buyer must create: regulatory pathway determination and submissions.

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-003', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'value of critical parameter lumen inner diameter [r m]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-001'}
- {'unknown': 'value of critical parameter lumen length [r m]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-002'}
- {'unknown': 'value of critical parameter surface roughness [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-003'}
- {'unknown': 'value of critical parameter operating pressure head [dP Pa]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-004'}
- {'unknown': 'value of critical parameter flow rate range [Q m^3/s]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-005'}
- {'unknown': 'value of critical parameter viscosity of flowed medium [mu Pa.s]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-006'}
- {'unknown': 'value of critical parameter occlusion growth tolerance [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-007'}
- {'unknown': 'acceptance criterion for verification VF-001', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-001'}

- {'unknown': 'acceptance criterion for verification VF-002', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-002'}
- {'unknown': 'acceptance criterion for verification VF-003', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-003'}
- {'unknown': 'acceptance criterion for verification VF-004', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-004'}
- {'unknown': 'acceptance criterion for verification VF-005', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-005'}
- {'unknown': 'acceptance criterion for verification VF-006', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-006'}
- {'unknown': 'acceptance criterion for verification VF-007', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-007'}
- {'unknown': 'acceptance criterion for verification VF-008', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-008'}

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	first physical article	implementing the proposed physical verification	performing physical verification	ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline	Article 001 (to be quoted)	1000
WP-02	constraint-compliance envelope article	per method	constraint-compliance envelope article	ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline	Article 002 (to be quoted)	1000
WP-03	article for VF-001	per method	Perform flow rate measurement	ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline	Article 003 (to be quoted)	1000
WP-04	article for VF-002	per method	run killer experiment	ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline	Article 004 (to be quoted)	1000
WP-05	article for VF-003	per method	bench flow-loop characterization	ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline	Article 005 (to be quoted)	1000
WP-06	article for VF-004	per method	pressure-drop mapping	ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline	Article 006 (to be quoted)	1000
WP-07	article for VF-005	per method	delivered-flow trend	ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline	Article 007 (to be quoted)	1000
WP-08	article for VF-006	per method	kink/collapse test across flow range	ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline	Article 008 (to be quoted)	1000
WP-09	article for VF-007	per method	pressure-flow mapping	ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline	Article 009 (to be quoted)	1000
WP-10	article for VF-008	per method	reverse-leakage rate	ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline	Article 010 (to be quoted)	1000

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 3 governing equations and a 10-step build plan. The next experiment is clearly specified. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.4557178723356231, EIG-per-cost 0.4557178723356231; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.4557178723356231, EIG-per-cost 0.4557178723356231; the falsification test carries the pre-registered pass/fail before any transfer)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.